SOUMIK SHANTO MONDOL

PhD Student | Electrical and Computer Engineering | Concordia University

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Website | Google Scholar | LinkedIn | GitHub

RESEARCH PROFILE

Doctoral researcher in medical ultrasound imaging, with interests in quantitative ultrasound, biomedical signal processing, inverse problems, optimization, and machine learning for medical imaging. Published research includes ultrasound attenuation estimation and multimodal breast-lesion analysis using explainable AI.

EDUCATION

PhD in Electrical and Computer Engineering | **Concordia University, Montréal · May 2026–Present**

Supervisor: Prof. Hassan Rivaz · IMPACT Lab

BSc in Electrical and Electronic Engineering | **BUET, Dhaka · Apr 2019–Jul 2024**

Major: Communication and Signal Processing · CGPA: 3.22/4.00

Thesis: *Fusion of Ultrasound Multi-Modal Images for Segmentation Assisted Classification of Breast Lesion*

RESEARCH EXPERIENCE

Doctoral Researcher | **IMPACT Lab, Concordia University · May 2026–Present**

- Conducting doctoral research in quantitative ultrasound and medical imaging.
- Presented a constrained-optimization approach to ultrasound attenuation estimation at IEEE IUS 2026.

Research Assistant | **Bangladesh University of Engineering and Technology (BUET) · Jul 2024–Nov 2025**

- Developed breast-lesion analysis methods combining B-mode ultrasound, H-scan and Nakagami imaging through cross-attention-based deep learning.
- Applied quantitative-ultrasound-informed learning and explainable AI to evaluate breast-lesion classification.
- Led experiments, quantitative evaluation and manuscript preparation for a first-author paper in *Physics in Medicine & Biology*.

SELECTED PUBLIC RESEARCH CONTRIBUTION

Constrained 2D ADMM for Pulse-Echo Attenuation Estimation | **IEEE IUS / QUASOS · 2026**

Developed and evaluated a constrained two-dimensional optimization approach for pulse-echo ultrasound attenuation estimation; presented at the IEEE International Ultrasonics Symposium QUASOS Challenge.

ADDITIONAL PROFESSIONAL EXPERIENCE

Intern Engineer | **Grameenphone Ltd., Dhaka · Dec 2023–Jan 2024**

Practical exposure to telecommunications engineering, network operations, and professional engineering workflows.

PUBLICATIONS & PRESENTATIONS

Journal article

S. S. Mondol and M. K. Hasan, "Enhancing B-mode-based breast cancer diagnosis via cross-attention fusion of H-scan and Nakagami imaging with multi-CAM-QUS-driven XAI," *Physics in Medicine & Biology*, vol. 70, no. 17, article 175011, 2025. DOI

Conference paper

M. S. A. Rafi, M. T. Rahman, M. A. Rahman, **S. S. Mondol**, M. M. Islam, and M. Ferdoushi, "Fault Detection Using Machine Learning in the Southwest Part of Bangladesh Power System," *2022 25th International Conference on Computer and Information Technology (ICCIT)*, 2022. DOI

Conference presentation

S. S. Mondol, A. K. Z. Tehrani, and H. Rivaz, "Constrained 2D ADMM for Robust Attenuation Estimation in Pulse-Echo Quantitative Ultrasound," IEEE International Ultrasonics Symposium (IUS), QUASOS Challenge poster presentation, 2026.

TECHNICAL SKILLS

Programming: Python, MATLAB, C, C++, Verilog
Machine learning: PyTorch, TensorFlow, deep learning, multimodal learning, explainable AI
Medical imaging & signal processing: Ultrasound RF processing, quantitative ultrasound, image analysis, spectral estimation, beamforming
Computational methods: Inverse problems, numerical optimization, ADMM, statistical analysis
Research tools: Linux, Git, Jupyter, Visual Studio Code, LaTeX

HONORS & AWARDS

- **International Tuition Award of Excellence** — Concordia University, doctoral admission
- **Government Merit Scholarship** — Bangladesh, 2019
- **6th Place** — National Chemistry Olympiad, Bangladesh, 2018
- **Champion** — Divisional Zoology Olympiad, Bangladesh, 2018

RESEARCH TRAINING

Deep Learning for Medical Imaging Summer School — École de technologie supérieure (ÉTS), Montréal · June 2026

SELECTED EARLIER ENGINEERING PROJECTS

- Single-channel speech separation using a Conv-TasNet-inspired temporal convolutional network.
- Two-dimensional ball-balancing system using PID control and image-based feedback.
- Embedded systems, robotics, and computer-network design and simulation.